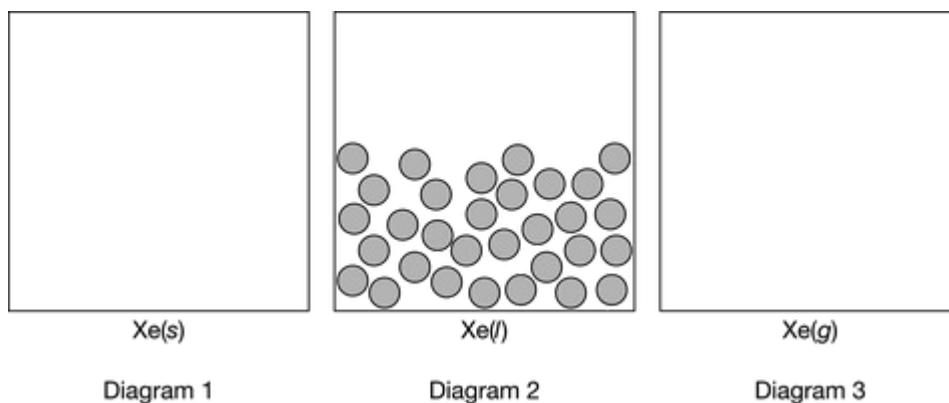


Unit 3 Progress Check: FRQ

Name _____

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to **Xe**.



(a) Diagram 2 shown above represents a particle-level view of liquid **Xe** atoms.

- (i) In the box in diagram 1, draw a particle-level view that represents a vessel that contains only solid **Xe**.



Please respond on separate paper, following directions from your teacher.

- (ii) In the box in diagram 2, draw a particle-level view that represents a vessel containing only gaseous **Xe**.



Please respond on separate paper, following directions from your teacher.

- (b) Let P and V represent the pressure and volume of the **Xe(g)** in the container in diagram 3. If a piston is used to reduce the volume of the gas to $\frac{V}{2}$ at a constant temperature, what is the new pressure in the container in terms of the original pressure, P ?



Please respond on separate paper, following directions from your teacher.

- (c) Describe what happens to the average speed of the **Xe** atoms in the container in diagram 3 as the



Unit 3 Progress Check: FRQ

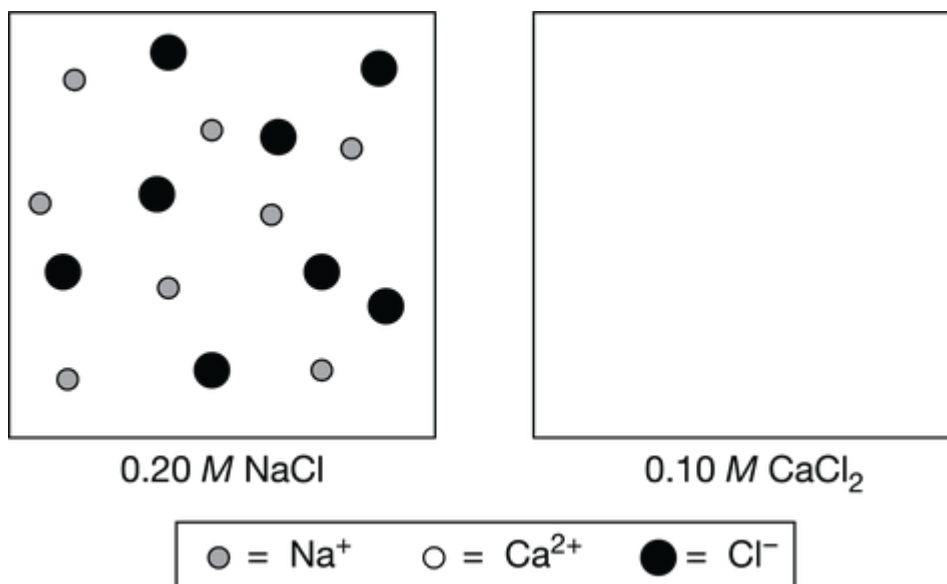
original volume V is reduced to $\frac{V}{2}$ at a constant temperature. Explain.



Please respond on separate paper, following directions from your teacher.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about solutions of NaCl and CaCl_2 .



- (a) Ions in a certain volume of $0.20 M \text{NaCl}(aq)$ are represented in the box above on the left. In the box above on the right, draw a particulate diagram that represents the ions in the same volume of $0.10 M \text{CaCl}_2(aq)$.



Please respond on separate paper, following directions from your teacher.

- (b) Briefly explain how the presence of charged ions in ionic solids such as $\text{NaCl}(s)$ helps to explain why $\text{NaCl}(s)$ is soluble in water.



Unit 3 Progress Check: FRQ



Please respond on separate paper, following directions from your teacher.

(c) What volume of **0.20 M NaCl(aq)** contains **10.0 g** of **NaCl** (molar mass **58 g/mol**)?



Please respond on separate paper, following directions from your teacher.